Oral Microbial Profile Discriminates Breast-fed From Formula-fed Infants

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ABSTRACT

Objectives: Little is known about the effect of diet on the oral microbiota of infants, although diet is known to affect the gut microbiota. The aims of the present study were to compare the oral microbiota in breast-fed and formula-fed infants, and investigate growth inhibition of streptococci by infant-isolated lactobacilli.

Methods: A total of 207 mothers consented to participation of their 3-month-old infants. A total of 146 (70.5%) infants were exclusively and 38 (18.4%) partially breast-fed, and 23 (11.1%) were exclusively formula-fed. Saliva from all of their infants was cultured for *Lactobacillus* species, with isolate identifications from 21 infants. *Lactobacillus* isolates were tested for their ability to suppress *Streptococcus mutans* and *S sanguinis.* Oral swabs from 73 infants were analysed by the Human Oral Microbe Identification Microarray (HOMIM) and by quantitative polymerase chain reaction for *Lactobacillus gasseri.*

Results: Lactobacilli were cultured from 27.8% of exclusively and partially breast-fed infants, but not from formula-fed infants. The prevalence of 14 HOMIM-detected taxa, and total salivary lactobacilli counts differed by feeding method. Multivariate modelling of HOMIM-detected bacteria and possible confounders clustered samples from breast-fed infants separately from formula-fed infants. The microbiota of breast-fed infants differed based on vaginal or C-section delivery. Isolates of *L plantarum, L gasseri*, and *L vaginalis* inhibited growth of the cariogenic *S mutans* and the commensal *S sanguinis: L plantarum >L gasseri >L vaginalis.*

Conclusions: The microbiota of the mouth differs between 3-month-old breast-fed and formula-fed infants. Possible mechanisms for microbial differences observed include species suppression by lactobacilli indigenous to breast milk.

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The microbiota in the oral cavity and other parts of the gastrointestinal (GI) tract develop from sterility at birth into the most heavily colonised parts of the human body. Different environmental conditions lead to distinct bacterial communities at the different anatomical niches, with >700 taxa identified in the mouth and 700 taxa identified in the colon, with minimal species overlap between the 2 sites, and significantly fewer taxa detected in the small intestine (1–4).

Establishment of stable bacterial ecosystems in the GI tract, including the mouth, evolves during a period of microbial variation chiefly during the first 2 years of life (1,4–6). In general, initial colonisers of facultatively anaerobic genera, such as streptococci and actinomyces, are succeeded by more strictly anaerobic genera, such as bifidobacteria in the gut and veillonellae in the mouth (4–9). Molecular-based studies have facilitated simultaneous detection of multiple bacterial species within the human microbiome, and findings suggest that changes during the period of variability and the final composition of the gut flora are related to negative health outcomes, such as allergy, obesity, and intestinal diseases in childhood (10–13), and myocardial infarction in adults (14). There are fewer studies examining the outcomes of early microbial colonisation in the mouth, but early acquisition of the cariogenic species *Streptococcus mutans* was associated with increased risk of dental caries (15).

The sources of bacteria transmitted to infants during the period of microbial variation include the mother's vaginal, gut and oral microbiota, the skin of caretakers and siblings, and breast milk and other foods. Microbial colonisation of the gut is influenced both by delivery method and by early feeding practices (4). Two recent publications report that the oral microbiota also differs by mode of delivery with detection of *Slackia exigua* only in caesarean section–delivered infants (16,17), and by delayed establishment of *S mutans* in vaginally compared with caesarean section–delivered infants (18). There is little information, however, about the effect of breast-feeding on the composition of the oral microbiota, although studies focusing on a limited number of taxa support the concept that breast-feeding can affect early bacterial colonisation in the mouth, consistent with reports for the gut (19,20).

Breast milk provides nutrition for the infant and is a source of lactobacilli, bifidobacteria, and streptococci (21,22). Furthermore, breast milk components inhibit growth and attachment of bacteria, such as the caries pathogen *S mutans* (23–25), but promote attachment of other *Streptococcus* and *Actinomyces* species (Johansson et al, unpublished data). Therefore, breast milk likely affects establishment of the microbiota in the mouth as well as in the gut.

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The aim of the present study was to compare the oral microbiota in 3-month-old breast-fed and formula-fed infants using the 16S rRNA probe Human Oral Microbe Identification Microarray (HOMIM, 26) and quantitative polymerase chain reaction (qPCR) to *Lactobacillus gasseri.* Total viable lactobacilli in the infant saliva were also compared between feeding groups, and the inhibitory capacity of isolated lactobacilli on growth of selected *Streptococcus* species was examined as a possible model of bacterial modulation of species colonisation.

METHODS

Study Group

The present study used a previously described cohort of 3-month-old infants (n = 207) (16). Information on feeding mode of the infant (exclusive or partial breast-feeding, or exclusive formula feeding), mode of delivery, gestational age at birth, and other possible confounders, such as infant health (allergy, infections, stomach discomfort), use of antibiotics or products containing probiotic bacteria, use of pacifier, and presence of teeth was obtained by a questionnaire. Information on infant body weight and length at birth and at 3 months of age and use of antibiotics at delivery was obtained from child health care and medical records. Oral samples were obtained from the infants at 3 months of age, between 1 and 3 hours after the latest meal (mean 2 hours) by an experienced dentist (P.L.H.) at a neighbourhood dental clinic. The study was approved by the regional ethical review board, Umeå, Sweden, and participating mothers signed informed consent at recruitment.

Oral Microbiota by 16S rDNA Probes in HOMIM

The mucosa of the cheeks, the tongue, and alveolar ridges of the infants were swabbed carefully using sterile cotton swabs (Applimed SA, Chatel-St-Denis, Switzerland). This mucosa-adherent biofilm was immediately transferred to Eppendorf tubes (Sarstedt, Nümbrecht, Germany) with 200-mL TE-buffer (Tris-EDTA, 10 mmol/Tris, 1 mmol/L ethylenediaminetetraacetic acid, pH 7.6), and stored at −80°C.

DNA was purified from samples from 55 breast-fed infants (of 184) that were randomly selected after stratification by mode of delivery and from 23 (all available) formula-fed infants as described (16). The DNA yield was 60 to 1220 ng DNA. Five samples were excluded because of low DNA yield, leaving 52 samples from breast-fed infants (44 exclusively and 8 partially breast-fed) and 21 samples from formula-fed infants for microarray analysis.

Purified DNA was assayed using 422 oligonucleotide probes to the 16S rRNA gene targeting >300 bacterial taxa in the HOMIM (*http://mim.forsyth.org/*) and by qPCR to *L gasseri* (27). Samples were analysed at the HOMIM facility at the Forsyth Institute (Cambridge, MA), as previously described (26). Hybridisation signals were read on a 6-level scale (0–5), with a lower limit of detection of approximately 10^4 cells (28).

Selective Culture of Salivary Lactobacilli and Characterisation of Isolates

Approximately 1 mL of whole saliva was collected into ice-chilled test tubes, transported, and cultured as described (16). Saliva samples from all of the infants (n = 207) were plated on Rogosa agar (Merck, Darmstadt, Germany; (29)) for *Lactobacillus* counts and on blood agar (Columbia base agar, supplemented with 5% horse blood) for total viable counts. The plates were incubated anaerobically for 48 to 72 hours. Counts were expressed as colony-forming unit per millilitre saliva.

For the last 21 infants sampled with cultivable *Lactobacillus* in saliva, on average 6 colonies were subcultured for identification. Isolates were characterised and typed by colony morphology, restriction fragment length polymorphism (RFLP) pattern, and carbohydrate fermentation tests (API 50 CH system, BioMérieux SA, Marcy-l'Etoile, France). Twenty-four isolates with differing characteristics were identified by comparing 16S rRNA gene sequences with the databases HOMD (*http://www.homd.org*) and NCBI (*http://blast.ncbi.nlm.nih.gov/Blast.cgi*).

RFLP

A 1420-bp fragment of the 16S rRNA gene was amplified using the primers 16S-LF and 16S-R (5′-AGA GTT TGA TCC TGG CTC AG-3′ and 5′-GGG CGG TGT GTA CAA GGC-3′, respectively) (30). Briefly, a 5-μL bacteria suspension in Milli-Q water (Millipore, Billerica, MA) was mixed with 25-μL HotStarTaq Mastermix (Qiagen, Hamburg, Germany), 16-μL Milli-Q water, and 20 pmol of the16S-LF and 16S-R primers in a total volume of 50 μL. The thermal cycling conditions were: 15 minutes at 95°C; 35 cycles of 30 seconds at 94°C, 30 seconds at 59°C, 1.5 minutes at 72°C; followed by 10 minutes at 72°C. An aliquot (10 μL) of the16S PCR product was digested for 1.5 hours at 37°C using the enzymes *Hinf*I and *Msl*I (New England BioLabs Inc, Ipswich, MA) in a total volume of 15 μL. The digest was applied on a 4% TBE (Tris/Borate/EDTA) NuSieve 3:1 agarose gel (Lonza, Rockland, ME) and products were visualised by ultraviolet light after ethidium bromide staining.

16S rDNA Sequencing

Aliquots of the 16S rDNA PCR products were purified and sequenced (Eurofins MWG Operon, Ebensberg, Germany), and sequences were compared with HOMD (*http://www.homd.org*) and NCBI (*http://blast.ncbi.nlm.nih.gov/Blast.cgi*) databases.

qPCR Detection of *L gasseri*

qPCR for *L gasseri* was performed as described by Byun et al (27) on DNA extracted from 190 of the mucosal swab samples, including samples from 63 infants with microarray data. The qPCR standard curve reaction mixture for *L gasseri* contained a total volume of 20 μL, consisting of Roche (Atlanta, GA) SYBR Green master mix 2X (10 μL), 20 μmol/L of each primer (0.25 μL), PCR grade water (5.5 μL), and *L gasseri* ATCC 29601 genomic DNA ranging from 2.5 ng/μL to 2.5 fg/μL (4 μL). The qPCR conditions were an initial denaturation at 95°C for 10 minutes, followed by 40 cycles at 95°C for 15 seconds, 59°C for 20 seconds, and 72°C for 30 seconds, and generated a 322-bp amplicon. Amplification and detection of DNA were performed using a Roche Lightcycler 480.

Growth Inhibition by Isolated *Lactobacillus* Species

The ability to inhibit growth of *S mutans* (strains Ingbritt and NG8) and *S sanguinis* (strain SK162) was tested for 1 randomly selected infant isolate each of *L plantarum, L vaginalis,* and *L gasseri* (identified by 16S rRNA gene sequences*)*, and corresponding type strains (*L plantarum* CCUG30503, *L vaginalis* CCUG31452, *L gasseri* CCUG31451 (CCUG, University of Göteborg, Sweden) in an agar overlay method using De Man-Rogosa-Sharpe agar (pH 5.0; Fluka, Buchs, Switzerland) in the bottom layer and M17 agar

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(pH 7.2; Fluka) in the top layer (31). Inhibition was tested for lactobacilli concentrations from 10^3 to 10^9 CFU/mL. *L reuteri* ATCC55730 (BioGaia, Stockholm, Sweden), previously shown to inhibit *S mutans* growth (32), was used as a positive control and agar plates without lactobacilli were a negative control. Growth after inhibition was scored as 0 = no visible colonies (complete inhibition), 1 = slightly reduced growth, or 2 = no growth inhibition (31). All of the experiments were done in triplicate and repeated 3 times.

Statistical Analysis

Data handling and statistical analyses were performed using SPSS version 18 software (SPSS Inc, Chicago, IL). Numbers of lactobacilli in saliva were logarithmically transformed to improve normality. Other measures were acceptably normal distributed. Bacterial counts, and length and weight at birth and at 3 months were averaged among infants by feeding method. Differences between means were tested using analysis of variance or Student *t* test. Data from the microarray were analysed as detected (all signal levels ≥ 1) or not detected (score 0) by child. Differences in prevalence distribution between groups were tested with a χ^2 test. Odds ratios to have a species/cluster detected in the microarray were estimated by logistic regression, with the possible confounder mode-of-delivery included as a covariate. Adjusting for number of teeth (three formula-fed infants had 1–3 erupting lower incisors) did not affect results further (data not shown). Differences for descriptive data are considered statistically significant for $P < 0.01$. The False Discovery Rate (33) was used to identify a *P* value accounting for multiple comparisons in the microarray. Thus, $P < 0.005$ was considered statistically significant for comparisons of the prevalence by microarray.

Multivariate partial least squares (PLS) analysis when 3 groups or PLS discriminant analysis (PLS-DA) when two groups modelling (SIMCA P+, version 12.0, Umetrics AB, Umeå, Sweden) was performed as previously described (34,35). In contrast to traditional regression models, PLS modelling, which defines the maximum separation between class members (here feeding method) in the data, is suitable for data where the number of observations is smaller than the number of variables, and where the X variables covary. Dichotomous HOMIM signals, and the selected individual characteristics, sex, weight, and length at birth and 3 months, gestational weeks at delivery, delivery mode, age at screening in days, use of pacifier, and presence of teeth built the X-block, and mode of feeding the Y-block (outcome). Variables were autoscaled to unit variance, and cross-validated prediction of Y was calculated (36). Cross-validation was done by a systematic prediction of one-seventh of the data by the remaining six-sevenths of the data. The separation of subjects is displayed in a score-loading plot and the importance of each x-variable by PLS loadings in a column plot. R^2 and Q^2 values give the capacity of the x-variables to explain (R^2) and predict (Q^2) the outcome.

Four different models were run: model 1 all 73 HOMIM-analysed samples (n = 73, comprising exclusively breast-fed, partially breast-fed, and exclusively formula-fed infants); model 2 exclusively breast- or formula-fed infants (n = 65); model 3 (n = 35); and model 4 (n = 30) as model 2 but restricted to vaginally and caesarean section–delivered infants, respectively. The same model groups but restricted to infants born vaginally in or after gestational week 37 were also tested.

RESULTS

Cohort Description

The study population comprised 207 infants of whom 146 (70.5%) were exclusively breast-fed, 38 (18.4%) partially breast-fed, and 23 (11.1%) exclusively formula-fed at the time of the sampling visit. In accordance with the inclusion criteria, all of the infants were healthy. Two infants received antibiotics immediately after birth, and none thereafter. These 2 infants were not included in the microarray subgroup. Except for vitamin D supplement (recommended for all infants), no child had received any medication. Approximately half of the mothers who had delivered by caesarean section with an acute complication had received intravenous antibiotics, but none of the mothers who delivered infants vaginally had antibiotics at delivery. None of the infants had been given products (including infant formula) with probiotics.

Among all of the infants (n = 207), there were no differences between infants grouped by mode of feeding in sex, length, and weight at birth and at 3 months of age, mode of delivery (80.2% were delivered vaginally), or being full term (98% were born in gestational week 37–42 and 2 infants were born in gestational week 35 [both exclusively breast-fed]) (Table 1). Although not statistically significant, exclusively formula-fed infants were on average 17 days older than breast-fed infants at the time of sampling. Among the children with samples analysed by microarray, the proportion of vaginally delivered infants was lower among breast-fed than among formula-fed infants. This was taken into account in the statistical analyses.

As described earlier, no significant differences were observed between the 207 participating infants and those infants whose mother declined to participate (16).

Saliva Culture Data

Saliva was evaluated for cultivable lactobacilli and total viable counts from all breast-fed and formula-fed infants (n = 207). Viable *Lactobacillus* species were detected in saliva from 32.4% and 27.0% of fully and partially breast-fed infants, respectively, but in none of the formula-fed infants (Table 1). Geometric means for infants with detectable lactobacilli (fully and partially breast-fed infants, respectively) were 7990 and 1862 CFU/mL saliva, and medians (max-min) 7450 (44–1,490,000) and 800 (100–72,000) CFU/mL saliva, respectively ($P = 0.005$). In breast-fed infants, the number of colony-forming units of lactobacilli per millilitre saliva correlated inversely with age in days ($r = -0.447$, $P = 0.003$). Among the 24 isolates (from 21 infants) with differing RFLP and API50 patterns, and 1 420-bp 16S rDNA sequences, *L gasseri* was most prevalent (87.5% of all of the isolates), but *L plantarum* (2 colonies) and *L vaginalis* (1 colony) were also identified.

qPCR Detection of *L gasseri*

L gasseri was detected in 25.3% of the oral swabs using qPCR (Table 1), including 63.0% mucosal swab samples that were positive for *Lactobacillus* species by selective culture of saliva ($P = 0.000$). The odds ratio for detecting *L gasseri* by qPCR in oral swabs if culture positive for lactotobacilli in saliva was 7.98 (95% CI 3.82–16.70, $P = 0.000$). When stratified by feeding method, 23.0% of fully and 38.9% of partially breast-fed, and 13.3% of formula-fed infants were positive for *L gasseri* by qPCR (Table 1). Similar proportions of samples were *L gasseri*–positive by qPCR in the subset of samples analysed by microarray data only (Table 1).

HOMIM-detected Bacteria and Clustering of Infants by Feeding Mode

Bacterial profiles of 52 exclusively or partly breast-fed, and 21 formula-fed infants were analysed. Overall, 81 bacterial species

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TABLE 1. Characteristics of participating 3-month-old infants by feeding mode

	Feeding mode at 3 months of age			
All study infants	Exclusively breast-fed, n = 146	Partly breast-fed, n = 38	Exclusively formula-fed, n = 23	*P*
Sex, % boys*	50.0	39.5	47.8	0.512
Age, months, mean (range)†	3.5 (2.4–5.5)	3.7 (2.4–5.3)	3.9 (2.8–5.1)	0.468
Vaginal delivery, % positive*	82.2	76.3	73.9	0.523
Weight, g†				
At birth	3504 (3414–3594)	3486 (3279–3694)	3549 (3299–3800)	0.913
At 3 months of age	6177 (6045–6309)	6043 (5770–6316)	6738 (5706–6369)	0.548
Length, cm†				
At birth	50.2 (49.8–50.6)	49.9 (49.1–50.7)	50.0 (49.0–51.0)	0.741
At 3 months of age	60.8 (60.3–61.2)	60.7 (59.9–61.6)	60.8 (59.8–61,9)	0.993
Microbiology at 3 mo of age				
Total viable count, mean (range $\times 10^{10}$)†	7.54 (7.47–7.61)	7.55 (7.39–7.72)	7.59 (7.39–7.78)	0.879
Total viable *Lactobacillus* count, % positive saliva samples*	32.4	27.0	0.0	0.006
L gasseri by qPCR, % positive swab samples*	23.0‡	38.9§	13.3‖	0.080
HOMIM-selected infants (n)	**n = 44**	**n = 8**	**n = 21**	***P***
Sex, % boys*	56.0	25.0	47.6	0.487
Age, mo, mean (range)†	3.4 (3.2–3.5)	3.4 (2.7–3.9)	3.7 (3.5–4.0)	0.07
Vaginal delivery, % positive*	43.2	0.0	76.2	0.001
Weight, g†				
At birth	3480 (3284–3677)	3256 (2642–3870)	3633 (3392–3873)	0.337
At 3 mo of age	6259 (5998–6540)	5787 (5080–6494)	6054 (5699–6401)	0.308
Length, cm†				
At birth	50.3 (49.4–51.1)	49.5 (46.3–50.7)	50.2 (49.2–51.2)	0.202
At 3 month of age	60.5 (59.7–61.3)	59.7 (57.3–62.1)	61.0 (59.9–62.1)	0.506
Microbiology at 3 months of age				
Total viable count, mean (range $\times 10^{10}$)†	7.55 (7.42–7.68)	7.91 (7.53–8.30)	7.63 (7.43–7.83)	0.09
Total viable *Lactobacillus* count, % positive saliva samples*	27.3	0.0	0.0	0.001¶
L gasseri by qPCR, % positive swab samples*	19.5#	37.5**	14.3††	0.207¶

HOMIM = Human Oral Microbe Identification Microarray; qPCR = quantitative polymerase chain reaction.
* Differences between group numbers tested with χ^2 test.
† Differences between group means tested with analysis of variance (3 groups) or *t* test (2 groups). If not stated otherwise, data are presented as mean with 95% confidence interval. Total bacteria are expressed as $\log^{10}$ values.
‡ 139/146 samples analysed.
§ 36/38 samples analysed.
‖ 15/23 samples analysed.
¶ One-sided test based on the association in all of the infants.
41/44 samples analysed.
** 8/8 samples analysed.
†† 14/21 samples analysed.

and 12 bacterial clusters (probes detecting ≥3 closely related species) in 7 phyla or divisions were detected by the HOMIM in oral biofilms of 3-month-old infants (supplemental Table S1, *http://links.lww.com/MPG/A168* [detection frequency {%} of taxa identified by HOMIM in exclusively formula or exclusively breast-fed, 3-month-old infants; taxa are identified by species name or by their Human Oral Taxon {HOT; *www.homd.org*} number if unnamed, or by cluster group; clusters are indicated by superscript numbers with targeted taxa in the footnote]). *Streptococcus* cluster III, *S anginosus/intermedius, S oralis/Streptococcus* sp HOT 064, *S parasanguinis* I and II, *Gemella haemolysans*, and *Veillonella atypica/parvula* were detected in >90% of samples, whereas other species were detected in low prevalence. The mean number of species/clusters detected per sample was 22 with a range of 10 to 41. The correlation between number of positive species/clusters and age was 0.259 ($P = 0.038$).

Data from the 73 infants were clustered using multivariate PLS modelling with feeding mode as outcome, and dichotomised

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HOMIM signals, salivary lactobacilli, and individual characteristics (see Statistical Analysis) for the x-variable block (model 1). This produced a model with 1 significant component and an explanatory and predictive capacity of 62.0% ($R^2 = 0.620$) and 55.7% ($Q^2 = 0.557$), respectively. Samples from exclusively breast-fed infants clustered separately from exclusively formula-fed infants, whereas partially breast-fed infants were interspersed among the breast-fed and formula-fed clusters (data not shown). Restricting analysis to exclusively breast-fed and exclusively formula-fed infants (model 2, n = 65) virtually separated the 2 different exclusive feeding method groups by their oral microbiota (Fig. 1A). Further analyses were restricted to comparisons of the oral microbiota of exclusively breast-fed (n = 44) and exclusively formula-fed infants (n = 21) (models 2–4).

Species Diversity in Breast-fed Compared With Formula-fed Infants

By the microarray, 79 species/clusters were detected in oral samples of exclusively breast-fed infants compared with 68 species/clusters from formula-fed infants ($P = 0.071$). Of the 93 species/clusters detected, 25 were detected only in breast-fed infants, compared with 14 species/clusters that were detected only in formula-fed infants (supplemental Fig. S1, *http://links.lww.com/MPG/A173* [bilateral bar chart of the detection frequency {%} among 3-month-old exclusively breast-fed or exclusively formula-fed infants; to illustrate the difference between feeding mode, probes to bacterial species are ordered by order of reactivity detection in breast-fed infants with taxa detected more frequently in breast-fed to the top and more frequently in formula-fed infants below; taxa are identified by species, by their HOT number if unnamed, or by cluster group {see supplemental Table S1 for detailed information}]); however, the mean number of species detected by child was higher in formula- than in breast-fed infants (mean 27 [range 16–35] vs mean 21 [range 10–41]; $P < 0.005$). The mean number of detected species by child was also higher in formula-fed infants after stratifying for mode of delivery (data not shown).

Microbiota in Breast-fed Compared With Formula-fed Infants

Data from 14 probes (13 species and 1 cluster) differed significantly between exclusively breast-fed and formula-fed infants when results from the χ^2 test ($P < 0.005$) were combined with the results from the logistic regression adjusted for mode of delivery (Fig. 2).

Actinomyces gerencseriae and *S australis* were detected more frequently in exclusively breast-fed compared with formula-fed infants. *Prevotella melaninogenica/histicola* and *Prevotella* cluster IV, *Lautropia mirabilis*, *Neisseria gonorrhoea/polysaccharea*, TM7[G-1] sp HOT 347/349, *Kingella oralis*, *Kingella* and *Neisseria* species, *Granulicatella adiacens/elegans*, *Leptotrichia buccalis/goodfellowii/Sneathia sanguinegens*, *Solobacterium moorei*, *Haemophilus* sp HOT 035/036, and *Veillonella* HOT 180 were detected more frequently in formula-fed infants.

S australis was detected more frequently in breast-fed than formula-fed infants regardless of delivery mode (Table S1, *http://links.lww.com/MPG/A168*). Prevalence of *Solobacterium moorei*, *Leptotrichia buccalis/goodfellowii/Sneathia sanguinegens*, and *Haemophilus* sp HOT 035/036 was higher in formula-fed infants regardless of delivery mode. The detection frequency of *Granulicatella elegans*, however, differed between delivery groups with a higher prevalence among caesarean-delivered breast-fed infants, but a higher prevalence in vaginally delivered formula-fed infants.

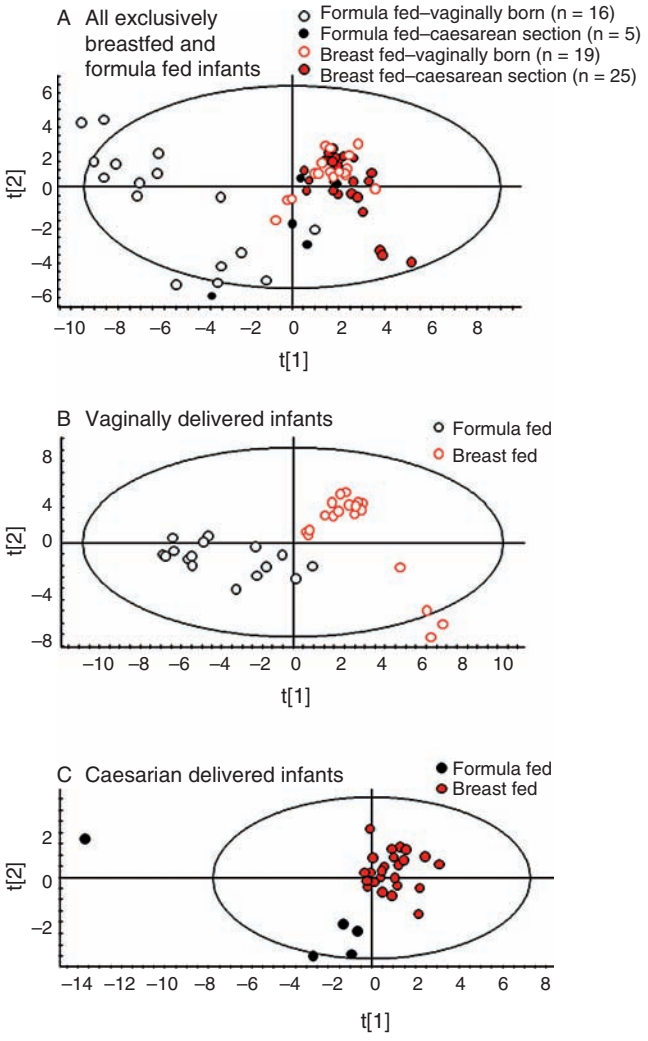


FIGURE 1. Partial least-squares discriminant analysis score scatterplots illustrating clustering of exclusively breast-fed (red symbols) compared with exclusively formula-fed (black symbols) 3-month-old infants by oral microbiota. Clustering was derived by multivariate partial least squares modelling using dichotomous Human Oral Microbe Identification Microarray (HOMIM) signals, lactobacilli cultured from saliva, and selected individual characteristics as the X-block and mode of feeding as the Y-block. A, Model 2 (all 65 exclusively breast- and formula-fed infant samples used for HOMIM microarray); B, model 3 (samples from vaginally delivered infants used for HOMIM microarray); and C, model 4 (samples from caesarean section–delivered infants used for HOMIM microarray). Red symbols with unfilled circles indicate vaginally and filled dots caesarean-delivered infants. Black symbols with unfilled circles indicate vaginally and filled dots caesarean-delivered infants.

PLS-DA modelling on mode of feeding in exclusively breast- and formula-fed infants (model 2, n = 65) rendered a model with 2 significant components, and explanatory and predictive capacities of 74.3% ($R^2 = 0.743$) and 59.8% ($Q^2 = 0.598$) (Fig. 1A, Fig. 3). Similarly, PLS-DA modelling on mode of feeding in vaginally delivered infants (model 3; n = 35) rendered a strong model

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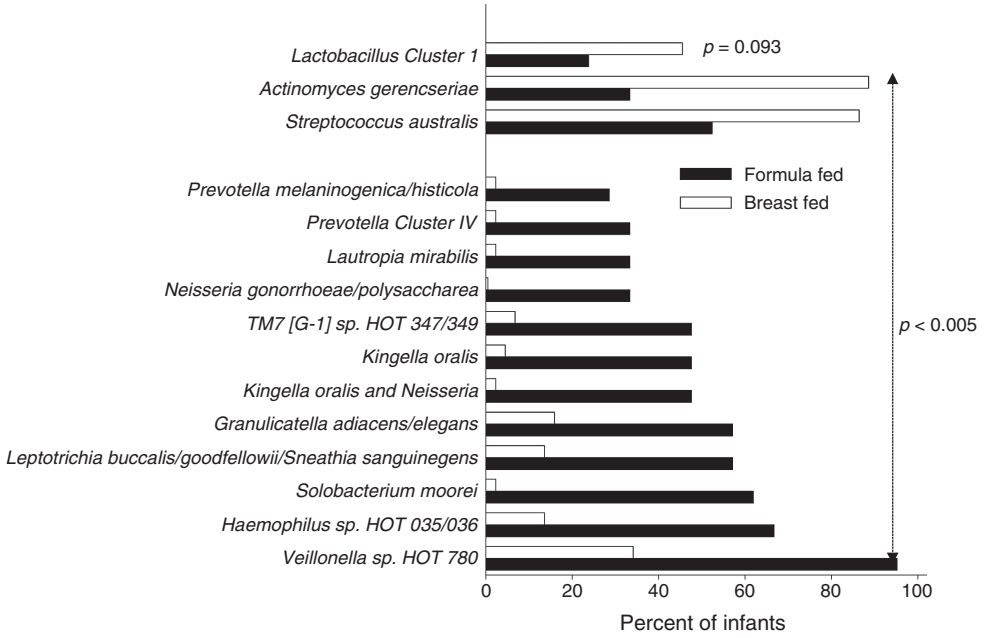


FIGURE 2. Major taxa differing between breast-fed and formula-fed infants. The figure illustrates Human Oral Microbe Identification Microarray detection frequency of 13 bacterial species and 1 cluster that differed by the false discovery rate, determined *P* value <0.005 and logistic regression including mode of delivery as a covariate. Prevalence of *Lactobacillus* cluster I is also shown.

($R^2 = 0.837$, $Q^2 = 0.717$), which completely separated breast-fed from formula-fed infants (Fig. 1B). The PLS-DA model on mode of feeding in caesarean section–delivered infants (model 4; n = 30) was weaker ($R^2 = 0.401$, $Q^2 = 0.185$, Fig. 1C). Models 2 and 3, involving the 2 infants born before gestational week 37, remained stable when infants born before gestational week 37 were excluded.

For both model 2 and 3, PLS-DA, which detects latent structures in the data swarm, confirmed the taxa that were significantly more prevalent in breast-fed infants by χ^2 testing and logistic

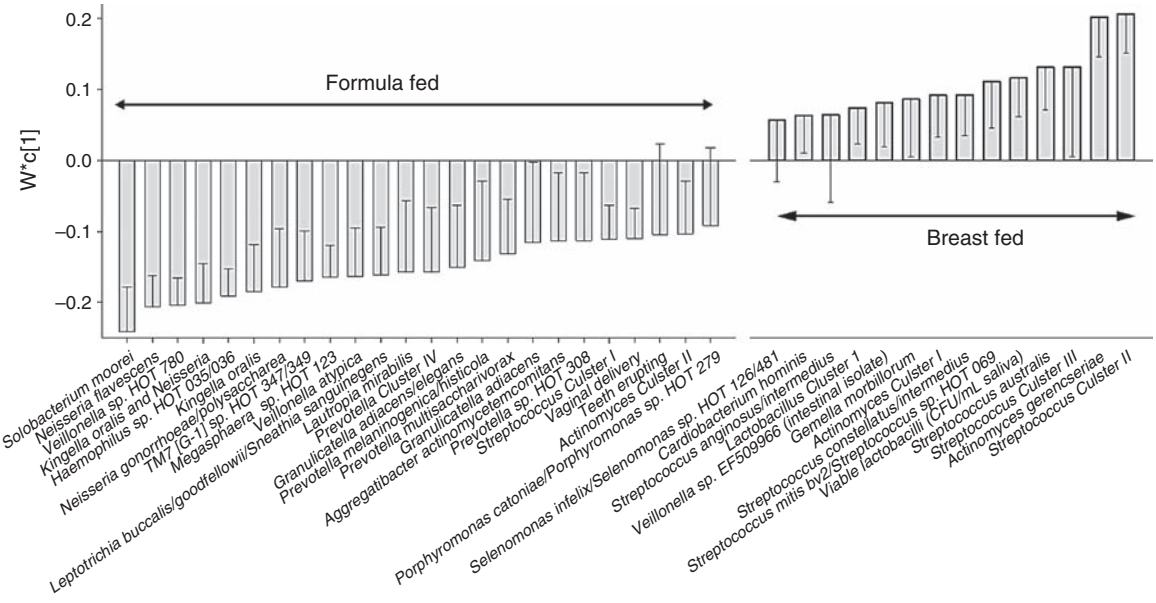


FIGURE 3. Taxa and some infant characteristics that differed between formula and breast-fed infants (model 2) by partial least-squares (PLS) modelling. The figure illustrates mean loadings with measurement error (95% confidence interval) in a PLS-derived column-loading plot for exclusively breast- versus formula-fed infants. The underlying PLS model used all of the bacterial taxa detected by microarray and potential confounders (see Statistical Analysis) simultaneously. Variables with highest loading in each group are displayed, and variables where the confidence interval does not pass 0 (zero) are statistically significant.

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regression, including cultivable lactobacilli. Model 2 (both vaginally and caesarean section–delivered infants) also revealed a breast-feeding association with *Streptococcus* clusters II and III, *S mitis* bv 2/*Streptococcus* sp HOT 069, *S constellatus/intermedius, Actinomyces* cluster I, *Gemella morbillorum*, *Veillonella* sp. EF509966 (intestinal isolate), *Lactobacillus* cluster I, and *Cardiobacterium hominis*, and model 3 (vaginally delivered infants only) the addition of *S anginosus/intermedius*, *Eubacterium yurii*, *Catonella morbi/Catonella* sp HOT 164, *Prevotella* cluster I, *Prevotella loescheii/Prevotella* sp HOT 472, and *Capnocytophaga granulosa/Capnocytophaga* sp HOT 326 (Table 2).

Similarly for both model 2 and 3, PLS-DA confirmed taxa that were significantly more prevalent in formula-fed infants by univariate testing. Model 2 also revealed a formula-feeding association with *Streptococcus* cluster I, *Granulicatella adiacens*, *Actinomyces* cluster II, *Prevotella* sp HOT 308, and *Aggregatibacter actinomycetemcomitans*, and in model 3 the addition of *S cristatus*, *Gemella haemolysans*, *Granulicatella elegans*, *Veillonella atypical/parvula*, and *Veillonella parvula* (Table 2).

Model 4 of caesarean section–delivered infants ($n = 30$) indicated that cultivable lactobacilli in saliva and *Slackia exigua* were significantly associated with being breast-fed, whereas *Campylobacter concisus/rectus* was associated with being formula-fed.

The PLS score scatterplot from model 2 showed distinct clusters (Fig. 1A) based on feeding and delivery methods. The groups of breast-fed, vaginally born infants; breast-fed, caesarean section–born infants; and 8 of the formula-fed, vaginally delivered infants each formed distinct clusters. The remaining formula-fed infants clustered with groups third and second.

Comparing microarray data and qPCR data, no statistically significant association was found between microarray detection of *Lactobacillus* cluster I and detection of *L gasseri* by qPCR ($P = 0.107$). Overall samples from 82% of the infants were negative for both *Lactobacillus* cluster I and *L gasseri* by qPCR, whereas 39% infants were positive for both *Lactobacillus* tests.

Saliva Lactobacilli From Breast-fed Infants With Inhibitory Growth Effects Against Streptococci

Isolates of the 3 identified *Lactobacillus* species (*L gasseri*, *L plantarum*, and *L vaginalis*) inhibited growth of *S mutans* (strains Ingbritt and NG8) and *S sanguinis* (strain SK162) (Fig. 4). *L plantarum* exhibited the strongest growth inhibition, followed by *L gasseri* and then *L vaginalis*. *L plantarum* totally inhibited growth at the lowest concentration of lactobacilli (10^3 CFU/mL), whereas *L gasseri* did so at concentrations $\geq 10^5$ CFU/mL. *L vaginalis* had no effect at concentrations $\leq 10^5$ CFU/mL and only partial inhibition effect at lactobacilli concentrations $\geq 10^7$.

DISCUSSION

In our investigation of the oral microbiota in 3-month-old infants, we sought differences associated with feeding method. We observed that microbial colonisation patterns clustered breast-fed infants separately from formula-fed infants. We further found that lactobacilli that inhibited growth of selected streptococci were cultured only in saliva from breast-fed infants.

In the present study, 93 species or species clusters of the approximately 300 taxa evaluated by the HOMIM were detected in the 3-month-old infants. The most prevalent species belonged to the phylum Firmicutes/Bacilli (genera *Streptococcus* and *Gemella*) and Firmicutes/Clostridia (genus *Veillonella*). Considering the detection threshold of about 10^4 cells for the HOMIM, species detected by the assay most likely represent bacteria colonising the mouth of the infants, rather than transient cells (26,37). Previous studies using bacterial culture have shown that oral bacterial colonisation in the neonate starts with streptococci from the viridians group (7,38), whereas significant colonisation by anaerobes was not detected before 2 months of age (7). The present frequent detection of species in Firmicutes phylum, and particularly within the genus *Streptococcus*, is consistent with these reports, as is detection of various anaerobes in lower frequencies in 3-month-old infants (8,38).

L gasseri was the most prevalent species detected among the 24 isolates from selective culture of saliva that were characterised by RFLP, sugar fermentation profile, and 16S rRNA gene sequencing. *L gasseri* was not detected in the microarray analysis of mucosal swab samples of the *L gasseri* qPCR-positive infants, despite a validated probe being included in the microarray. This difference likely reflected in part the increased sensitivity of the qPCR assay compared with the microarray assay, and in part that different samples were analysed. In contrast, lactobacilli in cluster I (*L casei*, *L paracasei*, *L rhamnosus*) were detected in microarray-analysed mucosal swabs. Because the microarray can be less sensitive than selective culture and most infants had $<10^4$ CFU/mL saliva of lactobacilli, qPCR was performed to evaluate whether the difference between saliva and microarray detection was because of method sensitivity or sample origin. qPCR confirmed that *L gasseri* were present in mucosal swabs, and that infants with salivary lactobacilli were 5 times more likely to have *L gasseri* detected in the mucosal swabs.

The present dataset is characterised by a larger number of variables than subjects, and by the presence of species that may be interdependent based on shared environmental needs or interspecies coaggregation. Under these conditions, the multivariate PLS method is suitable to search for subject clustering and identifying the variables characterizing the clusters.

The distribution of sex, weight, and length at birth and 3 months, number of siblings, and total salivary bacterial counts did not differ between infants selected for microarray analysis compared with all of the infants sampled. The major difference between breast-fed and formula-fed infants used for microarray analysis was that formula-fed infants were on average 17 days older, and that the prevalence of caesarean section, and the associated prevalence of complicated deliveries requiring use of intravenous antibiotics, was higher in breast-fed infants. This imbalance was the result of randomisation after stratification for delivery mode among the breast-fed infants, whereas all formula-fed infants were included to reach highest possible power. Intravenous use of antibiotics at delivery was previously shown to be noninfluential on the oral microbiota at 3 months of age (16). Furthermore, the PLS results from vaginally delivered infants were consistent with results from all of the infants. We, therefore, consider the groups of breast-fed and formula-fed infants suitable for PLS analysis, and the power trustworthy to discriminate between infants fed breast milk and those fed formula; however, confirmation of the results is needed, especially to exclude a possible systematic effect on presence of anaerobic species because of the slightly higher age in formula-fed infants, and the possibility of an unrevealed confounding by mode of delivery. In fact, the borderline significant correlation between age and mean number of species/clusters detected per child in the present study suggests that age may contribute to higher mean number of species/clusters detected per child in formula-fed infants.

Previously, the mode of delivery was found to affect the oral microbiota in infants at 3 (16) and 6 to 10 months of age (17), but any biological significance of the different colonisation patterns remains unproven (17). The present study indicates that at 3 months of age, and presumably later, the major effect of mode of delivery is in breast-fed infants. In the present analyses, infants being breast- or

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TABLE 2. Summary of bacteria associations with breast-feeding and formula feeding in univariate χ^2 testing (Fig. 3) and PLS modelling loading scores (for model 2, see Fig. 3)

	Association with breast-feeding			
		PLS-DA modelling		
Species	χ^2 test*	Model 2	Model 3	Model 4
Streptococcus cluster II	+	+	+	
Streptococcus cluster III	+	+	+	
Streptococcus anginosus/intermedius			+	
Streptococcus australis	+	+	+	
Streptococcus constellatus/intermedius		+	+	
Streptococcus mitis bv 2/*Streptococcus* sp HOT 069		+	+	
Gemella morbillorum		+		
Lactobacilli by cultivation	+	+	+	+
Lactobacillus cluster I (microarray)		+		
Eubacterium yurii			+	
Catonella morbi/Catonella sp HOT 164			+	
Veillonella sp EF509966†		+		
Actinomyces cluster I		+	+	
Actinomyces gerencseriae	+	+	+	
Slackia exigua				+
Prevotella cluster I			+	
Prevotella loescheii/Prevotella sp HOT 472			+	
Capnocytophaga granulosa/Capnocytophaga sp HOT 326			+	
Cardiobacterium hominis		+	+	
	Association with formula feeding			
		PLS-DA modelling		
Species	χ^2 test*	Model 2	Model 3	Model 4
Streptococcus cluster I		+	+	
Streptococcus cristatus			+	
Gemella haemolysans			+	
Granulicatella adiacens/elegans	+	+	+	
Granulicatella adiacens		+	+	
Granulicatella elegans			+	
Megasphaera sp HOT 123	+	+	+	
Solobacterium moorei	+	+	+	
Veillonella atypical/parvula			+	
Veillonella atypical	+	+	+	
Veillonella parvula			+	
Veillonella sp HOT 780	+	+	+	
Actinomyces cluster II		+	+	
Leptotrichia buccalis/goodfellowii/ Sneathia sanguinegens	+	+	+	
Prevotella cluster IV	+	+	+	
Prevotella melaninogenica/histicola	+	+		
Prevotella multisaccharivorax	+	+	+	
Prevotella sp HOT 308		+	+	
Kingella oralis	+	+	+	
Kingella oralis and *Neisseria*	+	+	+	
Neisseria flavescens	+	+	+	
Neisseria gonorrhoeae/polysaccharea	+	+	+	
Lautropia mirabilis	+	+	+	
Aggregatibacter actinomycetemcomitans		+	+	
Haemophilus sp HOT 035/036	+	+	+	
Campylobacter concisus/rectus				+
TM7 [G-1] sp HOT 347/349	+	+	+	

All PLS models use the same y and x variables. Model 2 comprises exclusively breast-fed and exclusively formula-fed infants (n = 65), model 3 (n = 35), and model 4 (n = 30) are as model 2 but restricted to vaginally and caesarean section delivered infants, respectively. PLS-DA = partial least-squares discriminant analysis.

* Differences where $P < 0.005$ are indicated.

† Intestinal isolate.

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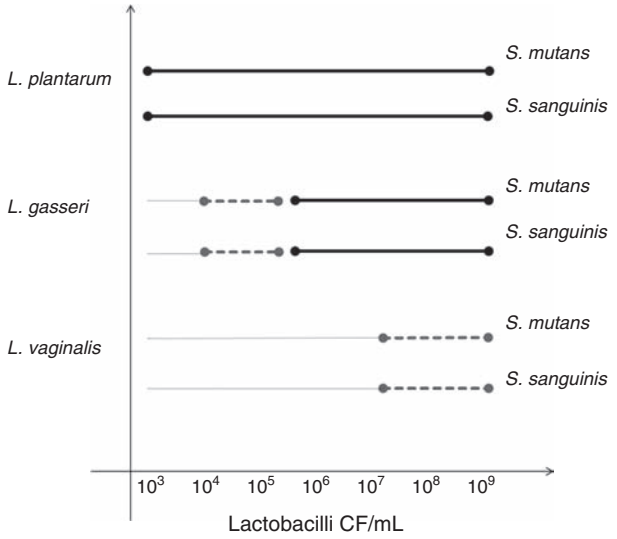


FIGURE 4. Growth inhibition by lactobacilli. Growth of *Streptococcus mutans* (strains Ingbritt and NG8) and *S sanguinis* (strain SK162) when exposed to increasing numbers of lactobacilli isolated from 3-month-old infants. Type strains gave identical results (data not shown). ●——● completely inhibited growth (score 0), ●– – –● partially inhibited growth (score 1), and ..... no effect on growth (score 2).

formula-fed and delivered by caesarean section were not separated in the PLS scatterplot. Although PLS theoretically allows modelling of few subjects, as in the subgroup formula-fed infants delivered by caesarean section (n = 5), results from the analyses separating out that group must be considered preliminary because we cannot distinguish how representative these 5 infants are for the group they represent. Hence, the results from this explorative study should be validated in an independent cohort, and mechanisms for observed effects explored.

In the first few months of life, the major influences on the establishment and succession of bacteria in the mouth comes from person-to-person transmission, composition of the infant's saliva, and microbial cross-talk (1,5,39), besides the presently shown effect of mode of feeding. Exposure to the mother's vaginal flora is considered to explain a greater diversity in the gut microbiota of vaginally born infants compared with those delivered by caesarean section (1,4,6,39), consistent with our recent report for the oral microbiota (16). In the present study, breast-fed infants had fewer species/clusters detected per child, whereas within the group of breast-fed infants, there were slightly more species/clusters found than among formula-fed infants; however, to evaluate a possible mechanism of the difference in species distribution between feeding groups, we observed that the microbiota of breast-fed infants compared with formula-fed infants included *Streptococcus* cluster I and II species (by prevalence) and *Lactobacillus* species (from PLS modelling); both genera that include health-associated species (40,41). In contrast, the biofilms of formula-fed infants included more anaerobic species usually associated with gingival inflammation (42) but not lactobacilli by culture of saliva. Although our observed detection of lactobacilli associated with breast-feeding is a novel and preliminary observation, the finding is supported by findings in another ongoing study in 4-month-old infants where viable lactobacilli are detected in saliva from approximately one-third of breast-fed infants but not in formula-fed infants. Possible mechanisms by which human milk affects the oral microbiota in early infancy include an interplay between milk receptors for protein–protein or lectin interactions with bacterial adhesins, innate immunity and immunoglobulin effects, and interspecies interactions by milk bacteria such as lactobacilli, bifidobacteria, and streptococci.

Although there is limited information on mechanisms for initial establishment of lactobacilli in the mouth of infants and the role of breast-milk as a source of oral bacteria (21,22), culturing lactobacilli only from breast-fed infants, and qPCR detection of *L gasseri* in breast-fed infants, in the present study is consistent with detection of lactobacilli in breast milk (43) and their transmission from the nipple and from milk (21,22,44). The finding that only a fraction of breast-fed infants have viable lactobacilli in saliva may reflect decreasing content of lactobacilli in milk by lactation age. Solis et al (22) found that bacterial levels in breast milk decreased from 10^5 at day 1 to $10–10^{3.7}$ CFU/mL at 3 months, and West et al (13) demonstrated that lactobacilli were no longer detected in feces with increasing age in breast-fed infants. This is in line with the present finding of a negative correlation between age in days and colony-forming unit of lactobacilli per millilitre of saliva. Another possible mechanism for differences in detection of lactobacilli may relate to avidity for lactobacilli to host receptors. Orally isolated lactobacilli adhere to saliva gp340 and mucins (45,46). These glycoproteins are present in polymorphic variants in saliva, and at least for *S mutans* adhesion to gp340 occurs in a polymorphism-dependent way (47).

Lactobacilli isolated from oral (31), breast milk (48), and other nonoral sites (31,49) inhibit growth of selected oral pathogens especially cariogenic mutans streptococci and *Candida albicans*. Growth inhibition of such species could be a mechanism for beneficial oral bacteria biofilm modulation, but will require testing against health-associated species, such as *S sanguinis* (50). A recent report indicated that 12-week daily exposure to lactobacilli did not affect the oral biofilm bacterial ecology as determined by a checkerboard DNA-DNA hybridisation method (51). In the present study, 3 *Lactobacillus* species *L plantarum*, *L gasseri*, and *L vaginalis,* isolated from the mouth of participating infants, inhibited growth of both *S mutans* and *S sanguinis,* demonstrating that the growth inhibitory capacity of oral lactobacilli in infants may not be specific to cariogenic (*S mutans*) or health-associated (*S sanguinis*) species. Prediction of the net effect of viable lactobacilli in saliva of breast-fed infants and the contribution of lactobacilli to the microbial community of breast-fed versus formula-fed infants, as well as later in life, remain to be explored.

In conclusion, the present findings demonstrate that the observed differences in the GI tract microbiota composition because of feeding mode extend to the oral cavity, and that viable lactobacilli detected in saliva from breast-fed, but not formula-fed infants, had an inhibitory effect on oral streptococci. A limitation of the present study is the low number of formula-fed infants, while a strength is the use of a microarray that facilitated simultaneous detection of a much greater number of oral taxa compared with culture methods, or single species or multiplex species-specific PCR. The finding that the oral microbiota differs between breast-fed and formula-fed infants, with a potentially more health-associated oral flora in breast-fed infants, indicates the value of prospective longitudinal studies where health outcomes, primarily early childhood caries but possibly also gut microflora and associated health conditions, are evaluated.

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